

Section : Physics

Q.1 Kirchhoff's second law is based on the law of conservation of

1. Momentum
2. Energy
3. Charge
4. Mass

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Two inductor coils of self inductance L_1 and L_2 having same length and same area of cross-section are placed so close that effective flux in one coil is completely linked with the other. The mutual inductance between them is, is equal to:

1. $L_1 L_2$
2. $\frac{L_1}{L_2}$
3. $(L_1 L_2)^2$
4. $\sqrt{L_1 L_2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.3 An object is placed at a distance of 10 cm from a concave mirror of focal length 15 cm. The magnification of the image is

1. -1.5
2. 1.5
3. 3
4. -3

Options 1. 1

2. 2
3. 3
4. 4

Q.4 A moving coil galvanometer has a coil of effective area A , number of turns N ,. The magnetic field B is radial. If current I is passed through the coil, the torque acting on the coil is:

1. NA^2BI
2. N^2A^2BI
3. $NABI$
4. NAB^2I

Options 1. 1

2. 2
3. 3
4. 4

Q.5 The core of a transformer is laminated, so as to

1. reduce the energy loss due to eddy currents
2. make it light weight
3. make it robust and strong
4. increase the secondary voltage

Options 1. 1

2. 2
3. 3
4. 4

Q.6 When 126 V is applied across a wire of 10 m length and 0.30 mm radius, the current density is $1.4 \times 10^4 \text{ A/m}^2$. What is the resistivity of the wire?

1. $1.2 \times 10^3 \Omega\text{m}$
2. $9 \times 10^{-5} \Omega\text{m}$
3. $9 \times 10^{-4} \Omega\text{m}$
4. $3 \times 10^{-8} \Omega\text{m}$

Options 1. 1

2. 2
3. 3

Q.7 A parallel combination of three resistors takes a current of 7.5 A from a 30 V battery. If the resistances of two resistors are $12\ \Omega$ and $10\ \Omega$, then the resistance of the third resistor is:

1. $5\ \Omega$
2. $10\ \Omega$
3. $15\ \Omega$
4. $20\ \Omega$

Options 1. 1
2. 2
3. 3
4. 4

Q.8 If the speeds of gamma rays, X-rays and microwaves are V_g , V_x and V_m , respectively in free space, then the correct relation between them is:

1. $V_g > V_x > V_m$
2. $V_g < V_x < V_m$
3. $V_g > V_m > V_x$
4. $V_g = V_x = V_m$

Options 1. 1
2. 2
3. 3
4. 4

Q.9 An electron, an α -particle, a proton and a neutron have the same kinetic energy. Which of these particles has the shortest de Broglie wavelength?

1. electron
2. proton
3. α -particle
4. neutron

Options 1. 1
2. 2
3. 3
4. 4

Q.10 A charge q is enclosed by a Gaussian spherical surface of radius R . If the radius is reduced to half of R , the outward electric flux will be

1. reduced to half of its initial value
2. the same as its initial value
3. twice its initial value
4. four times its initial value

Options 1. 1
2. 2
3. 3
4. 4

Q.11 A screen is placed at a distance of 90 cm from an object. The image of the object on the screen is formed by a convex lens placed at two different locations separated by 20 cm. The focal length of the lens is approximately

1. 22 cm
2. 20.8 cm
3. 21.4 cm
4. 19.2 cm

Options 1. 1
2. 2
3. 3
4. 4

Q.12 The electromagnetic radiation used to kill germs in water purifiers is

1. Gamma rays
2. X - rays
3. Infrared rays
4. Ultraviolet rays

Options 1. 1
2. 2
3. 3
4. 4

Q.13 Total internal reflection of a ray of light is possible when the ray goes from

1. denser to rarer medium and the angle of incidence is greater than the critical angle
2. denser to rarer medium and the angle of incidence is less than the critical angle
3. rarer to denser medium and the angle of incidence is greater than the critical angle
4. rarer to denser medium and the angle of incidence is less than the critical angle

Options 1. 1
2. 2
3. 3
4. 4

Q.14 A uniformly charged infinite plane sheet of charge density $2 \times 10^{-8}\ \text{C m}^{-2}$ is held in air.

What will be the separation between two equipotential surfaces in the region of electric field produced by the sheet, if the potential difference between them is 10 V? ($\epsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2\text{N}^{-1}\text{m}^{-2}$)

1. 8.85 m
2. 8.85 cm
3. 8.85 mm
4. 0.885 mm

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Two capacitors of capacitances C_1 and C_2 are connected in parallel.
If two-third of the total charge is shared by capacitor C_1 , the ratio of C_1 to C_2 is given by:

1. 1 : 2
2. 2 : 1
3. 2 : 3
4. 3 : 2

Options 1. 1

2. 2
3. 3
4. 4

Q.16 When a potential difference is applied across a copper wire, the drift velocity of the electrons is v . If the same potential difference is applied across another copper wire of the same length but twice the diameter, the drift velocity of electrons will be

1. $2v$
2. $3v$
3. v
4. $v/2$

Options 1. 1

2. 2
3. 3
4. 4

Q.17 A ray of light incident at 25° on one refracting surface of a prism of angle 60° suffers a deviation of 50° . What will be the angle of emergence?

1. 85°
2. 75°
3. 35°
4. 25°

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Two point charges of 1 C each are separated from each other by a distance of 1 m in vacuum.

The force of interaction between them is:

1. 3×10^9 N
2. 9×10^9 N
3. 9×10^8 N
4. 3×10^8 N

Options 1. 1

2. 2
3. 3
4. 4

Q.19 A solenoid of length 0.4 m having 500 turns of wire carries a current of 3 A.

A coil of 10 turns and radius 0.01 m carries a current of 0.4 A.

What will be the torque required to hold the coil in the middle of the solenoid with its axis perpendicular to the axis of the solenoid?

1. 4.9×10^{-7} N·m
2. 5.9×10^{-6} N·m
3. 3.75×10^{-7} N·m
4. 6.45×10^{-6} N·m

Options 1. 1

2. 2
3. 3
4. 4

Q.20 In a series LCR circuit, $R = 1 \text{ k}\Omega$, $C = 2 \mu\text{F}$ and the voltage across R is 100 V at resonant frequency $\omega = 200 \text{ rad/s}$.

At resonant frequency, the voltage across L is:

1. 40V
2. 250V
3. 2500V
4. 100V

Options 1. 1

2. 2
3. 3

- Q.21** The mass of the chlorine $^{35}_{17}\text{Cl}$ nucleus is 34.98 amu, which has protons of mass 17.133025 amu and neutrons of mass 18.155970 amu. The binding energy of the nucleus is nearly:
1. 287.82 MeV
 2. 270.0034 MeV
 3. 387.07 MeV
 4. 87.009845 MeV

Options

1. 1
2. 2
3. 3
4. 4

- Q.22** In a cyclotron, if a deuteron can gain a maximum energy of 40 MeV, then a proton can gain a maximum energy of
1. 20 MeV
 2. 40 MeV
 3. 60 MeV
 4. 80 MeV

Options

1. 1
2. 2
3. 3
4. 4

- Q.23** A uniformly charged sphere of $80\ \mu\text{C}$ and radius 2 cm is placed in air. The electric field intensity at a point 20 cm from the center of the sphere is:
1. $1.80 \times 10^7\ \text{N/C}$
 2. $1.80 \times 10^5\ \text{N/C}$
 3. $1.45 \times 10^5\ \text{N/C}$
 4. $1.45 \times 10^7\ \text{N/C}$

Options

1. 1
2. 2
3. 3
4. 4

- Q.24** An induced emf is produced when a magnet is plunged into a coil. The magnitude of the induced emf is independent of
1. the strength of the magnet
 2. the speed at which the magnet is moved
 3. the resistivity of the wire of the coil
 4. the number of turns in the coil

Options

1. 1
2. 2
3. 3
4. 4

- Q.25** A substance, when placed in a magnetic field acquires feeble magnetization in a direction opposite to the applied field, is
1. Alnico
 2. Bismuth
 3. Aluminium
 4. Sodium

Options

1. 1
2. 2
3. 3
4. 4

- Q.26** Some heat is developed in a uniform metallic wire across the ends of which a constant voltage V is applied. The heat developed will be tripled if
1. the radius of the wire is tripled
 2. the length of the wire is tripled
 3. both the radius and the length of the wire are reduced to one-third
 4. both the radius and the length of the wire are tripled

Options

1. 1
2. 2
3. 3
4. 4

- Q.27** Two positive point charges are 3 m apart in the air and the sum of the two charges is $20\ \mu\text{C}$. If the force between them is 0.075 N, the two charges are
1. 14 μC and 6 μC
 2. 4 μC and 16 μC
 3. 15 μC and 5 μC
 4. 12 μC and 8 μC

Options

1. 1
2. 2
3. 3
4. 4

- Q.28** A wire of length L meters, carrying a current I ampere is bent in the form of a circle. The magnitude of its magnetic moment, in SI units, is:

1. $\frac{IL^2}{\pi}$
2. $\frac{IL^2}{2\pi}$
3. $\frac{IL^2}{4\pi}$
4. $\frac{IL^2}{8\pi}$

Options 1. 1

2. 2
3. 3
4. 4

Q.29 The magnetic flux through a coil of resistance $6\ \Omega$, perpendicular to its plane, is varying according to $\phi = 6t^3 + 5t^2 + 4t - 6$ Wb. What will be the induced current through the coil at $t = 2$ s?

1. 60 A
2. 10 A
3. 16 A
4. 9 A

Options 1. 1

2. 2
3. 3
4. 4

Q.30 A slit is illuminated by light of wavelength $6500\ \text{\AA}$. The first minimum is observed at $\theta = 30^\circ$. The width of the slit is:

1. $13\ \mu\text{m}$
2. $3.25\ \mu\text{m}$
3. $1.3\ \mu\text{m}$
4. $1.7\ \mu\text{m}$

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The magnetic susceptibility of the material of a rod is 499. Permeability of vacuum is $4\pi \times 10^{-7}\ \text{H/m}$. The absolute permeability of the material of the rod is:

1. $4\pi \times 10^{-4}\ \text{H/m}$
2. $\pi \times 10^{-4}\ \text{H/m}$
3. $2\pi \times 10^{-4}\ \text{H/m}$
4. $3\pi \times 10^{-4}\ \text{H/m}$

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Which of the following statements is not true?
 1. Conductance is the reciprocal of resistance.
 2. Ohm's law is not applicable at very low and very high temperatures for conductors.
 3. Semiconductors obey Ohm's law.
 4. Ohm's law is not applicable to electron tubes, discharge tubes and electrolytes.

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Diodes are most commonly used for rectification of alternating voltages. This is because, in diode
 (A) current flows in one direction only
 (B) forward bias resistance is low as compared to reverse bias resistance
 (C) reverse current is independent of the voltage applied
 (D) reverse current is linearly dependent on the voltage applied
 Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (B) only
3. (A), (B) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.34 At equilibrium condition, the rate of generation of electron-hole pairs in Ge crystal

1. is more than rate of recombination of electron and hole pairs
2. is less than rate of recombination of electron and hole pairs
3. equals the rate of recombination of electron and hole pairs
4. is always zero

Options 1. 1

2. 2
3. 3

Q.35 A bar magnet held perpendicular to a uniform magnetic field experiences a torque T_1 .
By what angle should it be rotated to change the experienced torque to $0.5T_1$?

1. $\pi/6$
2. $\pi/4$
3. $\pi/3$
4. $\pi/2$

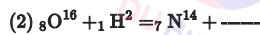
Options 1. 1
2. 2
3. 3
4. 4

Q.36 The sunlight is allowed to fall on a pinhole of diameter 10^{-4} mm and thus a diffraction pattern is obtained.
How will the size and intensity of the diffraction pattern be affected if the diameter of the pinhole is slightly increased?

1. size increases; intensity increases
2. size increases; intensity decreases
3. size decreases; intensity increases
4. size decreases; intensity decreases

Options 1. 1
2. 2
3. 3
4. 4

Q.37 The two possible nuclei to complete the reactions given below are, respectively



1. ${}_1\text{He}^4, {}_2\text{H}^1$
2. ${}_1\text{H}^1, {}_2\text{He}^4$
3. ${}_2\text{He}^2, {}_1\text{H}^2$
4. ${}_4\text{He}^2, {}_2\text{H}^2$

Options 1. 1
2. 2
3. 3
4. 4

Q.38 The electric field of a plane electromagnetic wave in a medium is given by
 $E = 100 \sin(300z - 6 \times 10^{10}t)$ V/m.

The velocity of the electromagnetic wave in this medium will be:

1. $0.5c$
2. $0.67c$
3. c
4. $1.5c$

Options 1. 1
2. 2
3. 3
4. 4

Q.39 The terminal voltages of a cell are found to be 29 V and 28 V when it delivers currents of 1 A and 2 A, respectively.
The emf and the internal resistance of the cell are, respectively:

1. 30 V; $2\ \Omega$
2. 30 V; $1\ \Omega$
3. 32 V; $2\ \Omega$
4. 32 V; $1\ \Omega$

Options 1. 1
2. 2
3. 3
4. 4

Q.40 A capacitor of capacitance $C = 10\ \mu\text{F}$ is charged by connecting it to a battery of emf 12 V.
The capacitor is now disconnected and reconnected to the battery with the polarity reversed.
The heat developed in the connecting wires will be:

1. 2.88 mJ
2. 1.44 mJ
3. 0.72 mJ
4. 0.36 mJ

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Which of the following statements is correct for the identification of a diode using a multimeter?

1. The multimeter shows an initial deflection which falls to zero
2. The multimeter displays current in one direction only
3. The multimeter displays current in both directions
4. Multimeter can not identify a diode

Options

1. 1
2. 2
3. 3
4. 4

Q.42 What is the de-Broglie wavelength associated with an electron, accelerated through a potential difference of 10 kV

1. 0.123 nm
2. 0.0123 nm
3. 12.3 nm
4. 1.23 nm

Options

1. 1
2. 2
3. 3
4. 4

Q.43 In a double slit experiment, if the distance between the slits be reduced to half, then the new fringe width will be

1. the same
2. doubled
3. halved
4. one-fourth

Options

1. 1
2. 2
3. 3
4. 4

Q.44 An alternating emf $\mathcal{E} = 220 \sin(50\pi t)$ is applied to an inductive circuit having self-inductance $L = \sqrt{2} \pi^{-1} \text{H}$. What will be the reading of an a.c. ammeter connected in the circuit?

1. 1.55 A
2. 2.2 A
3. 1.1 A
4. 4.4 A

Options

1. 1
2. 2
3. 3
4. 4

Q.45 A circular coil of 100 turns and area of cross-section 10^{-2} m^2 is rotated about its vertical diameter at 60 rpm in a uniform horizontal magnetic field of $3.5 \times 10^{-2} \text{ T}$. The maximum emf induced in the coil is:

1. 2.1 V
2. 13.3 V
3. 0.22 V
4. 0.37 V

Options

1. 1
2. 2
3. 3
4. 4

Q.46 An electric dipole is placed in a uniform electric field. Its potential energy will be minimum when the angle between its dipole moment and the electric field is

1. 0°
2. 90°
3. 180°
4. 30°

Options

1. 1
2. 2
3. 3
4. 4

Q.47 A radioactive sample decays $7/8$ times its initial quantity in 21 minutes. The half life of the sample will be

1. 7 min
2. 14 min
3. 63 min
4. 84 min

Options

1. 1
2. 2
3. 3
4. 4

Q.48 Match List-I with List-II

- | List-I | List-II |
|------------|---|
| Series | Values of n_1 and n_2 in Bohr's formula |
| (A) Lyman | (I) $n_1=4, n_2 \geq 5$ |
| (B) Balmer | (II) $n_1=3, n_2 \geq 4$ |

(C) Paschen (III) $n_1 = 2$, $n_2 \geq 3$

(D) Brackett (IV) $n_1 = 1$, $n_2 \geq 2$

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.49 An astronomical telescope has a large aperture to

1. reduce spherical aberration
2. have high resolution
3. reduce chromatic aberration
4. have low dispersion

Options

1. 1
2. 2
3. 3
4. 4

Q.50 Match List-I with List-II

List-I	List-II
Physical quantity	Unit
(A) Threshold frequency	(I) J
(B) Planck's constant	(II) V
(C) Work function	(III) 1/s
(D) Stopping potential	(IV) J s

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
2. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4